Proposal — Fort Frances & Area Wireless Broadband Solution

Prepared for: Jam21.net **Prepared by:** Northern Wireless Inc. (NWI) **Date issued:** July 3, 2026 **Reference:** NWI-PROP-FF-2026 · Rev 0

Executive Summary

Northern Wireless Inc. (NWI) proposes to engineer, supply, install and commission a complete fixed-wireless broadband network for Jam21.net serving Fort Frances and the surrounding area. The solution combines a 5.8 GHz wireless backbone with 802.11-based last-mile equipment to deliver high-speed internet to schools, businesses and residents — quickly, affordably, and with room to grow.

NWI has delivered wireless service to more than 70 customers, including municipal, policing and business clients, and has designed a proven set of products for rural deployments: a climate-controlled POP cabinet, a bandwidth-control unit, backhaul and backbone systems, and a low-cost client antenna. Customer premise equipment is now affordable enough (roughly \$200–\$300) for subscribers to own outright, removing the rental overhead that constrained earlier deployments.

On completion, Jam21.net owns and operates the entire network. NWI trains Jam21.net staff to maintain it.

Objective

Supply a complete wireless high-speed internet solution for Fort Frances and area — engineered, procured, installed and tested to a high standard using proven vendors — and ensure video-conferencing performs within the project scope.

Scope of Supply

NWI's turnkey scope covers:

- **Backbone:** 5.8 GHz point-to-point and point-to-multipoint links from a central tower.
- **Last mile:** 802.11 customer premise equipment (CPE) for residential and business subscribers; higher-capacity non-polling CPE available for video-conferencing and voice-over-IP users.
- **Head-end (POP):** antennas, amplifiers, AP enclosure, cabling and installation.
- **Core network:** bandwidth-control unit, main switch, primary and secondary servers, and T1/router hand-off, ready to integrate with Jam21.net's existing ISP services (RADIUS, email, web).
- **Premise equipment:** 100 initial subscriber connections (radio, antenna, cable, connectors).
- **Commissioning:** on-the-ground and installed on-site testing.

Boundaries of supply. Tower rental, IP address allocation, bandwidth purchase and resale, traffic/usage data collection, subscriber firewalls, and per-POP battery backup are Jam21.net's responsibility. NWI provides power conditioning and outage protection on all backbone equipment, and recommended spares.

Technical Solution

- **Capacity:** designed for **60 Mbps** against an estimated **50 Mbps** five-year area demand — headroom to extend service to nearby towns such as Atikokan.
- **Architecture:** an internet gateway is reached over a point-to-point wireless link; traffic is distributed across up to **8 break-out locations** feeding local last-mile cells. Existing water towers, communication towers or rooftops can host the antennas.
- **Subscriber connection:** a directional rooftop antenna connects by RF cable to the CPE radio, which bridges to the subscriber's computer over standard Ethernet — no drivers required. Each CPE authenticates by its unique MAC address; the bandwidth control unit assigns the subscriber's service level, adjustable at any time.
- **Service levels:** 0–60 Mbps, controlled by Jam21.net, at a typical planning ratio of 75 users per 1.5 Mbps.
- **Interoperability:** any Ethernet / TCP-IP device is supported (PC, Mac, most video-conferencing equipment). Standard internet security (HTTPS, VPN, PKI) is supported; each subscriber is responsible for securing its own traffic.

Project Approach

NWI manages the build using standard project-management disciplines:

- **Scope** — final equipment list and responsibility assignments confirmed once tower sites and the network design are approved; a work breakdown structure (WBS) drives procurement, cost and schedule.
- **Procurement** — components competitively sourced (typically 2–3 vendor quotes); orders placed on Jam21.net approval.
- **Cost** — hardware, software, installation and testing tracked against the approved budget.
- **Schedule** — a single master schedule integrates NWI, Jam21.net, Rainy River Futures Development Corporation, the chosen Telco and tower owners, with milestone and payment dates agreed before execution.
- **Quality** — only quality equipment procured; installations inspected and verified, with a deficiency punch-list closed out on an ongoing basis.
- **Risk** — key risks identified and mitigated before execution (see below).

Indicative Schedule

Milestone	Target date
Contract execution & kickoff	July 20, 2026
Site study, design & drawings approved	August 21, 2026
Procurement complete	September 18, 2026
Tower, POP & backbone installed	October 16, 2026
Last-mile & premise equipment installed	November 13, 2026
Commissioning, testing & handover	December 4, 2026

Dates are indicative and confirmed in the project master schedule after contract execution.

Pricing Summary

Firm, itemized turnkey pricing (supply and install, 100 subscriber connections):

Work package	Amount (CAD)
Engineering & Admin	$10,480
Tower	$17,000
Tower POP (head-end)	$5,500
Wireless Backhaul	$5,500
Core Wired Network	$19,500
Premise Equipment (100 subscribers)	$31,200
Commissioning & Testing	$8,320
Total supply & install	**$97,500**

Prices are in Canadian dollars and exclude applicable taxes. The full line-item breakdown is carried into the Supply & Install Contract. This proposal is valid for **60 days** from the date of issue.

Key Risks & Mitigation

Risk	Mitigation
Backbone link outage affecting multiple cells	On-hand spares and certified climbers; proven 5.8 GHz radios
Equipment failure at premise / last mile	Spare radios, RF cables and connectors held on-site
Schedule slippage across parties	Single agreed master schedule with milestone dates
Capacity growth	60 Mbps design vs. 50 Mbps five-year demand; per-user bandwidth control

Team & Qualifications

- **Jim Cox** — Design & Project Manager; ISP founder, wireless design, certified project-management specialist and scheduler.
- **Stephane Gauvin** — Engineering & Manufacturing; Certified Engineering Technologist (Instrumentation), designer of NWI's wireless cabinets and antennas.
- **Pat Cox** — RF installation and network administration; certified tower climber and satellite installer.
- **Partners** — Atikokan Enercom (tower / high-work installation), Proxim (wireless network equipment).

NWI's turnkey approach — design, install, commission, integrate and optimize — is the fastest, most reliable path to getting the network online on time and on budget.

Acceptance

To proceed, Jam21.net accepts this proposal and executes the accompanying **Supply & Install Contract** (Ref: NWI-CTR-FF-2026). Engineering and procurement begin on contract execution.

Contact: Jim Cox, Northern Wireless Inc. · jcox@nwon.com · 613-843-9244

Prepared by Northern Wireless Inc. — turnkey wireless internet solutions. Figures are illustrative for planning purposes.